

Database Management Homework #1
Due Date: 10/17/2024 (Thursday) 23:59
(Please turn in to 作業管理 of i學園Plus)

1. Consider a database design involving employees, insurance policies, and dependents.
 - (a) An employee has attributes **e-id**, **e-name** and **address**.
 - (b) A policy has attributes **id** and **cost**.
 - (c) A dependent has attributes **d-name** and **birthdate**.

The design wishes to express the following concepts:

- (1) A policy is owned by exactly one employee, but an employee can own zero, one or more policies.
- (2) Dependent is a **weak entity type**, and each dependent entity instance is uniquely identified by taking d-name in conjunction with e-id of an employee entity instance (which, intuitively, covers the given dependent).

Please draw an E-R diagram for this database **using binary relationships** that cover associating employee, policy and dependent. Specify identifiers in your entity types. (You must specify **maximum cardinality**, **minimum cardinality**)

2. Suppose the car dealer sells both new and used cars. Please draw an **E-R diagram** according to each of the following condition (your E-R diagram should include the **maximum and minimum cardinality**):
 - (1) A sales person may sell many cars but each car is sold by only one sales person. (Use **Salesperson** and **Car** as entities).
 - (2) A customer may buy many cars but each car is sold to only one customer. (Use **Customer** and **Car** as entities).
3. An Internet store sends email messages to customers and would like to use a database to keep track of which messages are sent to which customers. A **MESSAGE** has a message id (m-id), a subject (subject), and a body (body). A **CUSTOMER** is identified by the email address (email), and customer's data includes the attributes: name, sex, household-size, and address. When an email is sent, the date (send-date) is recorded. You can assume any reasonable domains for these attributes. Please draw an E-R diagram that completely describes the entities and relationships. (You must specify **maximum cardinality**, **minimum cardinality**)